MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 3: Topology. Differentiation of complex functions.

Nick Strigul
Washington State University Vancouver
nick.strigul@wsu.edu

January 23, 2023

Lecture outline

1. Elements of topology
2. Complex functions and derivatives
3. Differentiable functions
4. Properties of differentiable functions

Elements of topology: basic definitions

Definition

Let $z_0 \in \mathbb{C}$. Then

1. Given an $\epsilon > 0$ a *neighborhood* (ϵ neighborhood) of z_0 is the $\{z| \ |z - z_0| < \epsilon\}$.
2. z_0 is said to be a *limit point* of a set of points S if every neighborhood of z_0 contains a point of S distinct from z_0.
3. If a set S contains all of its limit points, then S is said to be *closed.*
4. A point $z \in S$ is called an *interior point of* S if there is a neighborhood of z that contains only points of S.
5. An *open set* consists only of interior points.
6. A point z is a *boundary point* of a set S if every neighborhood of z contains at least one point of S and at least one point not in S.
7. The set of boundary points of S is called the *boundary* of S.
8. The *closure* of a set S is the set consisting of S and all its limit points. It is denoted by $\bar{S}$.

Elements of topology: examples

Note: Every neighborhood of a limit point contains infinitely many points of S. A limit point of a set may or may not be in the set.

Example

$S = \{z \mid |z| < 1\}$. $z = 1$ is a limit point. In fact, any $z = a$, where $|a| = 1$ is a limit point. Furthermore, every point $z_0 \in S$ is a limit point of S. Note that S does not contain all of its limit points.

Example

The set $S = \{z \mid |z| \leq 1\}$ contains all of its limit points.

Example

$\{z \mid |z| < 1\}$ is not closed, whereas $\{z \mid |z| \leq 1\}$ is closed.

Example

The set $S = \{1, 2, \ldots, n, \ldots\}$is closed because the set of limit points of S is the empty set Φ and $\Phi \subset S$.

Elements of topology: closure of the set

Theorem

The closure $\bar{S}$ of any set S is closed.

Proof.

Suppose that z_0 is a limit point of $\bar{S}$. Then $\exists$ a sequence of points $z_1, z_2, \ldots \in \bar{S}$ which converges to z_0. If an infinite subsequence of this sequence lies in S, we are done since this implies z_0 is limit point of S and therefore is in $\bar{S}$.
If not, i.e., if this sequence $z_1, z_2, \ldots$ contains only finitely many points of S, remove this finite set of points. Then we have an infinite sequence of limit points of S which converge to z_0.
Given $\epsilon > 0$, consider $\epsilon/2$. $\exists$ a z_n such that $|z_0 - z_n| < \epsilon/2$

Now z_n is a limit point of S. Therefore $\exists$ $\tilde{z}_n \in S$ such that $|z_n - \tilde{z}_n| < \epsilon/2$. Hence

$$|z_0 - \tilde{z}_n| \leq |z_0 - z_n| + |z_n - \tilde{z}_n| < \epsilon$$

By definition z_0 is a limit point of S and hence $z_0 \in \bar{S}$.

$\square$

Elements of topology: connected and bounded sets

Definition

An set S is said to be *connected* if any two of its points can be joined by a polygonal arc (a continuous curve composed of finitely many straight-line segments) all of whose points belong to S.

Remark: Often a connected set definition is applied to only open sets.

Definition

A *domain* is an open connected set.

Definition

A *region* is a domain plus some of its boundary points.

Definition

A set S is *bounded* if $\exists$ a number $R > 0$ such that if $a + bi \in S$, then

$$a^2 + b^2 < R^2.$$

Elements of topology: sequences. Part 1

Definition

A complex *sequence* $\{z_n\}$ is an assignment of a complex number z_n to each positive integer n.

Remark: We often write $z_1, z_2, \ldots, z_n, \ldots$

Example

The same *point* in the complex plane may well correspond to several or even infinitely many distinct *terms* of a sequence. Thus the sequence

$$1, 0, 3, 0, 5, 0, 7, \ldots$$

has the unique limit point 0.

Elements of topology: sequences. Part 2

Example

The sequence

$$1, \frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}, \frac{1}{5}, \frac{4}{5}, \ldots$$

has the limit points 1 and 0, where the first belongs to the sequence and the second does not.

Definition

If L is a complex number, we say that $\{z_n\}$ *converges* to L if given $\epsilon > 0$ $\exists$ a positive integer N such that $|z_n - L| < \epsilon \quad \text{if } n > N$.
When $\{z_n\}$ converges to L, we write $\lim_{n\to\infty} z_n = L$ or $z_n \to L$. If a sequence does not converge, the we say that it *diverges.*

Complex functions and derivatives. Part 1.

Let $f(z)$ be a function of a complex variable which is defined and single-valued on a domain E, and let z_0 be any point of E. Then the *difference quotient*

$$\frac{f(z) - f(z_0)}{z - z_0} \tag{1}$$

is a function of z defined for any point $z \neq z_0$ of the set E. The limit of (1) as $z \to z_0$, $z \in E$, provided it exists, is called the *derivative* of the function $f(z)$ at the point z_0 relative to the set E, denoted by $f'_E(z_0)$ or simply $f'(z_0)$, and the function $f(z)$ itself is said to be *differentiable* at the point z_0 relative to the set E.

Complex functions and derivatives. Part 2.

In the special case where E is an interval (finite or infinite) of the real axis, $f(z)$ is a function of the real variable $z = x$ and takes values which are in general complex:

$$f(z) = f(x) = \phi(x) + i\psi(x).$$

If $\psi(x) = 0$, if the values of $f(z)$ are real, our definitions of a derivative and of differentiability reduce to the usual definitions given in elementary calculus. However, if $\psi(x) \neq 0$, then, writing

$$\frac{f(x) - f(x_0)}{x - x_0} = \frac{\phi(x) - \phi(x_0)}{x - x_0} + i\frac{\psi(x) - \psi(x_0)}{x - x_0},$$

we conclude that $f'(x_0)$ exists if and only if the derivatives $\phi'(x_0)$ and $\psi'(x_0)$ exist, and that

$$f'(x_0) = \phi'(x_0) + i\psi'(x_0).$$

Complex functions and derivatives. Example.

Example

If

$$f(x) = a\cos x + ib\sin x$$

(where x is any real number), then

$$f'(x) = -a\sin x + ib\cos x.$$

Introducing the notation $\Delta z = z - z_0$ for the increment of the independent variable and $\Delta_E f(z) = f(z) - f(z_0)$ for the increment of the function $f(z)$ (relative to the set E), we have the following necessary and sufficient condition for differentiability:

Differentiable functions. Part 1

Theorem

The function $f(z)$ is differentiable at the point $z_0 \in E$ (relative to the set E) if and only if $f(z)$ can be written in the form

$$\Delta_E f(z) = A\Delta z + \varepsilon(z, z_0)\Delta z, \tag{2}$$

where $\varepsilon(z, z_0) \to 0$ as $\Delta z \to 0$ ($z \in E$), and A is a constant independent of Δz and ε.

Proof.

If $f(z)$ has a derivative $f'_E(z_0)$ at z_0, then, by definition

$$\frac{\Delta_E f(z)}{\Delta z} = f'_E(z_0) + \varepsilon(z, z_0), \tag{3}$$

where $\varepsilon(z, z_0) \to 0$ as $\Delta z \to 0$. Multiplying (3) by Δz, we find that $\Delta_E f(z)$ can written in the form (2) with $A = f'_E(z_0)$. Conversely, if $\Delta_E f(z)$ can be written in the form (2), then dividing by Δz and taking the limit as $\Delta z \to 0$, we find that $f'_E(z_0)$ exists and equal A. $\square$

Differentiable functions. Part 2

Remark. It is an immediate consequence of (2) that is $f(z)$ is differentiable at the point $z_0 \in E$, it is also continuous at z_0 (relative to E).

By the *differential* of a function $f(z)$ which is differentiable at the point $z_0 \in E$, we mean the principal linear part of the increment $\Delta_E f(z)$, i.e., the quantity

$$d_E f(z_0) = f'_E(z_0)\Delta z = f'_E(z_0)dz, \tag{4}$$

where, as in elementary calculus, we set $dz = \Delta z$. We can then write

$$f'_E(z_0) = \frac{d_E f(z_0)}{dz}. \tag{5}$$

For simplicity, we usually omit the subscript E in the expression like (2) through (5), unless we want to emphasize the role of the set E.

Differentiable functions: example. Part 1

Example

To clarify the role of the set E relative to which the derivative is taken, let E be real axis and consider the function

$$f(z) = f(x) = x. \tag{6}$$

Obviously, the derivative $f'_E(z)$ exists for all $z = x \in E$, since

$$\frac{f(z) - f(z_0)}{z - z_0} = \frac{x - x_0}{x - x_0} = 1 \quad (z \in E).$$

We now extend the function $f(z)$ to the whole complex plane $\mathbb{C}$, by writing (6) for all $z = x + iy \in \mathbb{C}$. Then $f(z)$ is continuous for any $z \in \mathbb{C}$ and coincides with the original function when $z \in E$.

Differentiable functions: example. Part 2

Example

However, the difference quotient is now

$$\frac{f(z) - f(z_0)}{z - z_0} = \frac{x - x_0}{(x - x_0) + i(y - y_0)}, \tag{7}$$

which has no limit as $z \to z_0 \in \mathbb{C}$. In fact, (7) approaches 0 if $z \to z_0$ along the line $x = x_0$, whereas (7) approaches 1 if $z \to z_0$ along the line $y = y_0$. In other words, $f'_{\mathbb{C}}(z_0)$ does not exist for any $z \in \mathbb{C}$. Thus it is already apparent from this simple example that the requirement that $f(z)$ is differentiable regarded as a function of a complex variable $z = x + iy$ is much more stringent than the requirement that $f(z)$ be differentiable only for real values of z.

Properties of differentiable functions. Part 1

It follows from our definition of a derivative and the properties of limits that the basic differentiation rules familiar from elementary calculus can be extended to the case of functions of a complex variable. We now list some of these rules. In each case, $f(z), f_1(z), f_2(z)$, etc., is assumed to be differentiable at a given point $z \in E$ (relative to E).

Rule 1. If $f(z) = z$, then

$$\frac{df(z)}{dz} = 1,$$

whereas if $f(z) = \text{const}$, then

$$\frac{df(z)}{dz} = 0.$$

Loosely speaking, the derivative of the independent variable is 1, and the derivative of any constant is 0.

Properties of differentiable functions. Part 2

Rule 2. Differentiation of sums, products and powers. If c is a constant, then

$$\frac{d[cf(z)]}{dz} = c\frac{df(z)}{dz}.$$

Moreover, we have

$$\frac{d}{dz}[f_1(z) + f_2(z) + \cdots + f_n(z)] = \frac{df_1(z)}{dz} + \frac{df_2(z)}{dz} + \cdots + \frac{df_n(z)}{dz},$$

$$\frac{d}{dz}[f_1(z)f_2(z)\cdots f_n(z)] = \frac{df_1(z)}{dz}f_2(z)\cdots f_n(z) + f_1(z)\frac{df_2(z)}{dz} \cdots f_n(z) + f_1(z)f_2(z)\cdots\frac{df_n(z)}{dz},$$

$$\frac{d}{dz}[f(z)]^n = n[f(z)]^{n-1}f'(z).$$

In particular,

$$\frac{d}{dz}z^n = nz^{n-1}.$$

Properties of differentiable functions. Part 3

Rule 3. Differentiation of a quotient. If $f_2(z) \neq 0$, then

$$\frac{d}{dz}\left[\frac{f_1(z)}{f_2(z)}\right] = \frac{f_2(z)\frac{df_1(z)}{dz} - f_1(z)\frac{df_2(z)}{dz}}{[f_2(z)]^2}. \tag{8}$$

Properties of differentiable functions. Part 4

Rule 4. Differentiation of a composite function. Suppose the function $f(z)$ with domain E and range $\mathbb{C}$ is differentiable at the point $z_0 \in E$, and suppose $\mathbb{C}$ is an infinite set with $w_0 = f(z_0)$ as a limit point. Moreover, suppose the function $\phi(w)$ has domain $\mathbb{C}$ and is differentiable at the point $w_0 \in \mathbb{C}$. Then the *composite function*

$$\phi[f(z)]$$

is differentiable at the point z_0, and

$$\frac{d_E \phi[f(z_0)]}{dz} = \frac{d_{\mathbb{C}} \phi(w_0)}{dw} \frac{d_E f(z_0)}{dz} \tag{9}$$

Thank you